Name: ______________

Date: ______________

Periodic Functions & Modeling

Student learning objectives for this unit

SUPPORTING SKILLS LEGEND

I — Introduce | D — Deepen | A — Apply | R — Reinforce

☐ **1. I can graph basic sine and cosine functions using key features.**

RELATED SKILLS

- I — Locate quarter-period key points of a periodic function.
- D — Use tabular patterns to predict graph shape and behavior.
- A — Determine unit-circle coordinates for common angles.

☐ **2. I can identify amplitude, midline, period, and phase shift from an equation or graph.**

RELATED SKILLS

- I — Determine a sinusoid's midline and vertical shift.
- I — Determine amplitude and vertical reflection.
- I — Determine horizontal shift from a graph or equation.
- I — Determine period from a graph or equation.
- I — Recall and correctly use the terms amplitude, midline, period, phase shift, vertical shift, and asymptote.

☐ **3. I can graph transformed sine and cosine functions.**

RELATED SKILLS

- D — Distinguish stretches from compressions and horizontal effects from vertical effects.
- D — Locate quarter-period key points of a periodic function.
- A — Interpret transformations applied to function inputs.
- A — Interpret transformations applied to function outputs.

☐ **4. I can write sine and cosine equations that represent a periodic graph.**

RELATED SKILLS

- D — Connect parameters in a formula to features of its graph.
- D — Determine a sinusoid's midline and vertical shift.
- D — Determine amplitude and vertical reflection.
- D — Determine horizontal shift from a graph or equation.
- D — Determine period from a graph or equation.

☐ **5. I can construct a sinusoidal model from contextual information or data.**

RELATED SKILLS

- I — Translate contextual maximum, minimum, cycle, and starting position into model parameters.
- A — Determine a sinusoid's midline and vertical shift.
- A — Determine amplitude and vertical reflection.
- A — Determine period from a graph or equation.
- A — Restrict a model to meaningful contextual inputs.

☐ **6. I can interpret the parameters and outputs of a sinusoidal model in context.**

RELATED SKILLS

- D — Explain a model parameter in the language of its context.
- D — Translate contextual maximum, minimum, cycle, and starting position into model parameters.
- A — Attach and interpret units for rates, inputs, and outputs.
- A — Identify input and output values in a formula, graph, table, or context.

☐ **7. I can graph tangent, secant, and cosecant functions with their asymptotes.**

RELATED SKILLS

- I — Locate vertical asymptotes of tangent, secant, and cosecant.
- D — Relate sine/cosecant, cosine/secant, and tangent/cotangent.
- A — Interpret transformations applied to function inputs.
- A — Locate quarter-period key points of a periodic function.

☐ **8. I can determine whether a function is even, odd, or neither from its graph or formula.**

RELATED SKILLS

- D — Test a formula or graph for even or odd symmetry.
